G H Raisoni International Skill Tech University, Pune

End Semester Examinations
School of Engineering and Technology
FY B.Tech, Semester - I (All)

Course Name: Digital Electronics | **Course Code: 24BTCL203**

Max Marks: 60 | **Duration: 90 minutes**

Instructions:

1. You are permitted to use calculators during the exam.
2. 2. Utilize diagrams and equations where relevant, and always include units
3. 3.Students must label their answers with the corresponding question number, such as Q.No. 1A, Q. No. 2C, etc.
4. **Solve any two Questions from options A, B, and C. Option D is mandatory to solve in every question.**

Q. No.	Sub Q.No.	Solve any two Questions from options a, b, and c. Option d is mandatory	Marks	CO #	BL
1.	A	Define a digital signal. How do digital signals differ from analog signals?	5	CO1	L1
	B	Explain the operation of a NAND and NOR gates. Provide their truth table also.	5	CO1	L1
	C	Discuss the differences between one's complement and two's complement in binary arithmetic. Give example for bit operations also.	5	CO1	L1
	D	Perform the following conversions: i. $(1011)_2 = (\quad)_8$ ii. $(53)_{10} = (\quad)_2$	2	CO1	L1
		Solve any two Questions from options a, b, and c. Option d is mandatory			
2	A	Design and explain a half-adder circuit. Write the truth table and provide the logic gate diagram.	5	CO2	L2
	B	Use a K-map to simplify the Boolean function: F (A, B, C) = ∑ (1, 3, 6, 7). Realize the circuit using basic logic gates.	5	CO2	L2
	C	What is a multiplexer (MUX)? Explain its working and draw the truth table for a 4-to-1 multiplexer.	5	CO2	L2
	D	Define "Don't care" conditions in combinational logic circuits.	2	CO2	L2
		Solve any two Questions from options a, b, and c. Option d is mandatory			
3	A	What are flip-flops in digital electronics? Explain their working principle and major applications in digital circuits.	5	CO3	L3
	B	Discuss the operation of the S-R flip-flop. Provide its truth table and outline its applications.	5	CO3	L3
	C	Describe the operation of a 4-bit shift register with data flow	5	CO3	L3
	D	What is a counter in a sequential circuit?	2	CO3	L3

		Solve any two Questions from options a, b, and c. Option d is mandatory			
4	A	Highlight the differences between Analog-to-Digital Converters (ADC) and Digital-to-Analog Converters (DAC). List at least five distinctions.	5	CO4	L4
	B	How does a Weighted Resistor Digital-to-Analog Converter (DAC) work?	5	CO4	L4
	C	What key specifications are associated with Digital-to-Analog (D/A) converters?	5	CO4	L4
	D	What is a read and write operations involved in memory organization	2	CO4	L4
		Solve any two Questions from options a, b, and c. Option d is mandatory			
5	A	Explain the working principle of CMOS technology and its applications in digital systems.	5	CO5	L5
	B	Discuss the principles of ECL (Emitter-Coupled Logic) and its applications in digital electronics.	5	CO5	L5
	C	Compare ROM (Read-Only Memory) and RAM (Random Access Memory). Briefly explain the different types of RAM. (Type of I)	5	CO5	L5
	D	Explain the working principle of TTL in logic families.	2	CO5	L5

- End of the Question Paper -